

Unit 9, Lesson 9: Surface Area in the Real World



Benchmark

MA.912.GR.4.6 Solve mathematical and real-world problems involving the surface area of three-dimensional figures limited to cylinders, pyramids, prisms, cones, and spheres.



Learning Targets

- ☐ I can solve real-world problems involving the surface area of cylinders, pyramids, prisms, cones, and spheres.
- ☐ I can solve real-world problems involving the surface area of composite figures.



9.9.1 Warm-Up: Math Magic

Arthur Benjamin (b. 1961) is a professor of mathematics at Harvey Mudd College in California. Dr. Benjamin is often referred to as a “mathemagician” because he is known for his quick mental computational skills. Dr. Benjamin’s TED talks have been viewed millions of times on the internet.

Try one of Dr. Benjamin’s math “magic tricks.”

- Pick a number between 1 and 10.
- Double your number.
- Add 10 to your number.
- Divide by 2.
- Subtract the number you started with originally.

1. What number did you get?

2. Compare your number with your partner’s number. What do you notice?

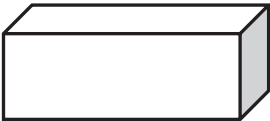
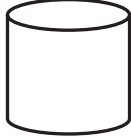
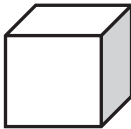
3. Why do you think the “trick” works?



9.9.2 Collaboration: The Best Package

Manuel is a graphic designer and works with various types of packaging. For the artwork, Manuel needs a container with a surface area of at least 250 square inches (in.) to ship the company's products.

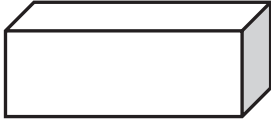
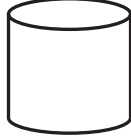
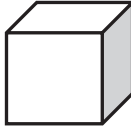
1. The three types of packaging that the company has in stock are shown in the table. Find the surface area for each type of packaging.

Packaging	Dimensions	Surface Area
Rectangular Prism 	l : 7 in. w : 10 in. h : 4 in.	
Cylinder 	d : 8 in. h : 8 in.	
Cube 	s : 12 in.	

2. Rank the packaging's surface area from smallest to largest.

Rank	Packaging
1 (smallest)	
2	
3 (largest)	

3. The prices for packaging are listed in the table. Based on the surface area of each package, find the cost of each type of packaging.

Packaging	Pricing	Cost of Packaging
Rectangular Prism 	\$1.75 per 100 square feet.	
Cylinder 	\$2.25 per 100 square feet.	
Cube 	\$1.25 per 100 square feet.	

4. If Manuel wants to use the cheapest packaging possible for his client, which packaging should he choose?



9.9.3 Guided Instruction: Real-Life Applications of Surface Area

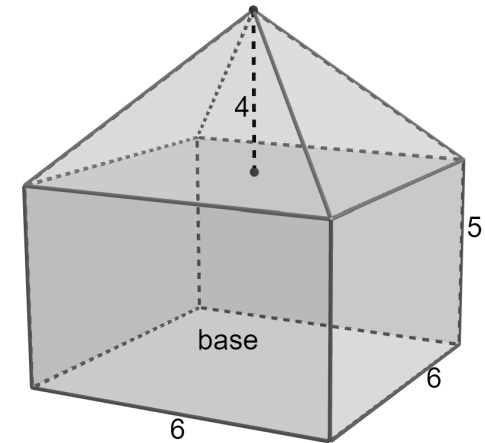
Jeni owns a bakery and is making two cakes.

Cake A	Cake B
Cake A is in the shape of a rectangular prism that is 3 inches thick, 9 inches long, and 12 inches wide.	The cake is in the shape of a cylinder and has a diameter of 8 inches and a height of 6 inches.

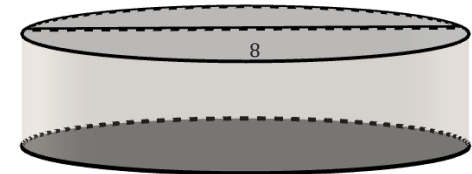
- When Jeni frosts the cake, she applies the frosting to all surfaces except for the bottom of the cake. Determine the amount of frosting Jeni needs to apply to each cake. Show your work.

Cake A	Cake B

- Jeni makes a two-layer cake, as shown, where the measurements are in inches (in.) She will frost all of the cake except for the base. The frosting between the pyramid and the prism helps to hold the two cakes together. Determine the total surface area that Jeni needs to frost.



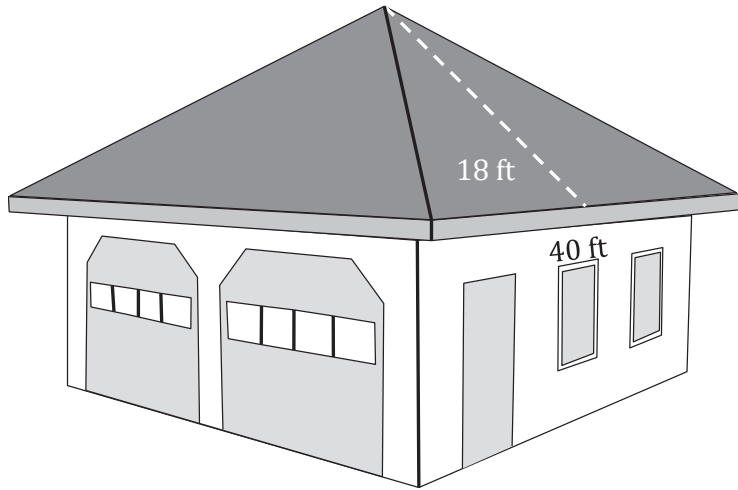
- Jeni's recipe for buttercream icing typically frosts an 8-inch round cake that is 1.5 inches high. How many batches of frosting will Jeni need to frost the two-layer cake?





9.9.4 Guided Practice: Real-Life Applications of Surface Area

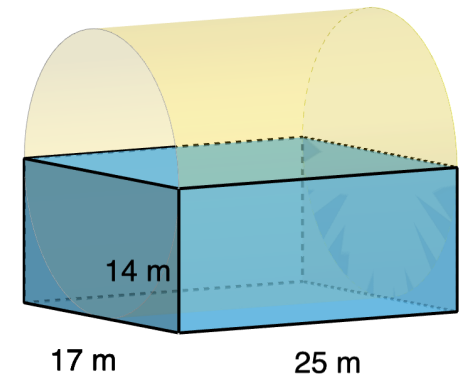
1. Lourdes' garage is in the shape of a square pyramid. The garage is shown, with dimensions given in feet (ft).



Lourdes is replacing the roof and installing shingles. In roofing terminology, a "square" is 100 square feet. Composition shingles come in bundles that cover $\frac{1}{3}$ of a square.

How many bundles should Lourdes buy to cover the roof?

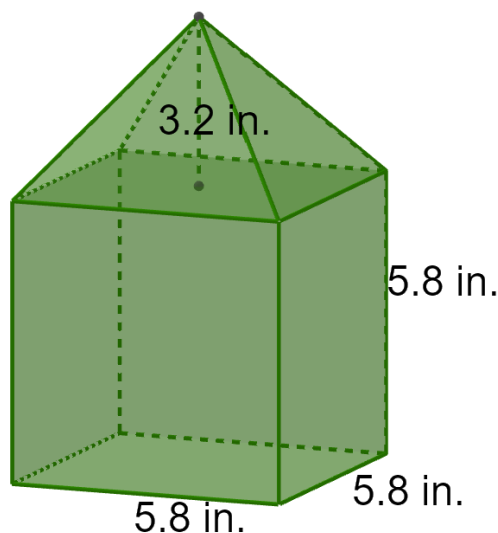
2. An architect is designing an art gallery. She used computer software to create a rough sketch of her idea. Her sketch is shown with measurements in meters (m). She would like for the art gallery to be all glass. To determine the estimated cost, she would need to find the area of all of the surfaces except for the floor. Determine the total surface area, in square meters, of the glass in the art gallery. Round to the nearest thousandth.





9.9.5 Wrap-Up: Ticket Out the Door

1. Jillian was building a birdhouse. Some of the dimensions, in inches (in.), of the birdhouse are shown. Determine the total amount of wood, in square inches, that Jillian will need to build the outside of the birdhouse.



Unit 9, Lesson 10: Determining How Dilations Affect Area



Benchmark

MA.912.GR.4.3 Extend previous understanding of scale drawings and scale factors to determine how dilations affect the area of two-dimensional figures and the surface area or volume of three-dimensional figures.



Learning Target

- ☐ I can discover the relationship between area and the scale factor of a dilation.